

Supplementary Information

Cell-free biosynthesis and engineering of ribosomally synthesized lanthipeptides

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Supplementary Tables

Supplementary Table 1 Gene sequences used in this study (genes are codon-optimized for *E. coli*).

Gene name	Sequence (5'-3')
<i>sboA</i>	atggcgaaagcagcaaatgaacctgtgtggagatcgaagcgatgaacagcctgcaggagctgacctggaggagctggacaacgtgctgggtgctgggtg gctgggtgtatccagaccattagccacgaatgccgtatgaacagctggcaattctgtttacctgctgcagc
<i>sboA_A-1R</i>	atggcgaaagcagcaaatgaacctgtgtggagatcgaagcgatgaacagcctgcaggagctgacctggaggagctggacaacgtgctgggtgctgggtg gctgggtgtatccagaccattagccacgaatgccgtatgaacagctggcaattctgtttacctgctgcagc
<i>sboM</i>	atggtgaaaaatggagaacaccgaaaaagaacaaactgttcaacaagtttaccgtaacttcacagaccaacttttagcgaatacgcataatgcaaaactgcga gacgagttctcggagacaaaagtctgtgatttatatgaacacctgatcctgctgattaacgaaaaacgtctgctgggtatctgaaaggcagcaccag cgaggaaactgtacgaatatttcaacgacgtgctgtgctggcaagtacatcattgatgagattgcgaacgttttccgaagctggtagccgtgacccg gctgaaaaactacgtgctgtatcgaagtgaggcaactgttctgaaagacgttccgaagctgacccagaacgtttcattagcaaccaatgcgagaa ctttaaaatggagacctgcagatcaccgtgagcggcgatttccacaacgtgtagcggcgtttgcatcctgttttacaagggtgaacgtgtgtattataagggc aaaaacaacttcgcgaaccaactgctgacgagatttttagcaagctgggtgtagtgcgaggaagcctggaaatttctgccgctgacatcgtactgcggc gattactattgggaggaatatacgaacacaccagcattcgaacagcaacatcagcgcgagcagtagtactataaacgtttcgggtacgtgctggcgtggc gtatctgtgaacatcagcagcctgcaatttgaacacattctggcgagcggcgacatcccgaactgggtgagcgttgagaccattttcaacctgcagccgta cgagattctgagcgaacatcgcggacaaaagatcgtgaagcgtaacgcggaagcgttctgctgaccggtctgctgcccgtggcgagcgaacg agggtgttcgggtgagataacgcggatcctgggtgcaaaattgttggcgcaaggtgctgttattcagaacaaaactgagcagatcagcgtgagcgtc agggtgttcgtaaaaaacaagagccacctgcccgttctacatcagcagggcgaggaacaatactttgaagttaaagactatctgcgtgagctgctgttc ggctttaagaaagtaccagctacgttgagaaccagaagaacaccttctgagcatcattgagaaatatagcgaaaagatcgacgttcgtattctgtttcgt acaccaaagattacagcatcgtgctgacccgtgctgtagcggcgaggtatagcgaacacatcattctgagaaggttcaggaacacgtggttaac taccaaaagcgtgaactgtgcgagagcgaatfaaacagctggagcaaatggacatcccgtatttcagcatgaagagcgaacgtgcgacgtgtacgac tgtatggtaacgcgggttggagcacctactatagcccgtacgaactgattcgtgaagaaatccagaacctgagcgtgcacatcattgaggaacaatacaaac tgatcgagtttagcattagcagcaccctgctgctgtataaccaggaagcgtgcccgtgattaacgacgttgaggataaaagctacagcgacagcattctgt ataaggtgtgtgcagaaacatcgtgtgctgctggcgcgatatttaccgtgagcaacagcgaccaaacctgcaactggctgacctgagcatcaacaacgt ggactgctggaactgagcccgtggtgacccctgtatgtggcgtgagcgtgtgtgctgctgacgcatgacccctgctggaggaaga ttatcgtaaagagtgctgtagcgttctgccgtacctgtataacagcattctgaagaatacagatgtatgaccaacgacagcttctacatgggtcgtatgggc gttctgtataccctgaagaaactgagcaagaagtgggcaaggacctggataactgggtgaacatcgagattgataacgttattagccgtctgaacatcgtt gacaccgatttctgaccgcatcagcagcgttatcccgcgatctgctggagaacctgaacaaagtacattttaggagactgagcgacatcatatataa gcaggcgaaagaaaaggataactcgtgttctggggtaacgagcaaaagcaacaacgtgagcctggcgacggtaacctgggcatcgaactggcgctgg tgcgtctgattaaagtcttgacaacagcaacaacaaggagaccatcagcaccctgctgaagaaagcgattaaacttcgacaacctgcagcgtctgaaca cggctgggtgggataaccgttaattttagccacagcgcgaactgggtgctgacccgagcgttctgctgagcgtgacacacagctgtaaca aacgataagaccgcgaccattagcagcaccgctgagcaagatgttattcagacagcgtcgaagatacaagaggtgggtctgaacctggcgaaactta gcctgtgccaggtgtgagcggcaacctgctgacctgggttcacaccagaactctttgacgaaaacaacatcgtgaacaactacaacctgaagaaattc gttcgtgagaacttctacaagatccacctgtatgtgtgctgcaaaacggctggaatgtgcaacttcaacaccaagtacgacagctatgcgctgtttaccggatca gcggcattctgtacgcgaccacctgtatctgaaaaacgacatcaagagcgatgtgctgctgcccgaactgtgagcctggttccgctggtgagctggagc caccgcgagttcgaaaaaggtggcagcgggtggcagctggagccaccgcaatttgagaagtaa
<i>sboT</i>	atgggcatgaaaaatcattctgcagaacaacgaagcgactgcctgctggcgtgctacgcgatgctgctgaacaacctgggtctggagatcccgtgtacg aaatttataaagaggacatgattccggcgatggtctgaactgagctacgtgatcaaaactgaacaagcgtttcaaaagtggcatcaaaagcgtatcgtattct gctgaaggacatcgcgggtgtgctgtgagaaacattccggttatcctgctggaacgataaccatttgggttctgctgacacacagctgaacaagaata taccatcattgatccggcgattgtgtaaaatcacctacagcgagggaagaggtgctgaagcactatagcgggtatttgcgtgaccgttttcaaaaggcgtggaatt aagcgtatcaaatacagcaacctgttcgtgagcagctgcgtcaaacattctgctgaagccggcgatgtattttattctggcgtgggtctgaccaggcg ggcgtgttctgttttagcatcaccattcgaacattctggacgttaaatacagcttcatcgcgagcctgctgtagctgatgagcgtggtgtgttgaactgatc agcgtgctgattaaagaaacgaacctcagcgagttcaacaacgactttgatgaaactacgttaacaagctgttcaaaatgctgctgaacaaaccgctgctg tattttcgtaatagagcaacggtgtgctggcgagaaaatcaacctgaagagcaccatccgtgatagcattaccctgaaactgctgccgagcatcgttgg ttcattagcctgtttgtgttattctgtacctgctgacctcagcacaagctgaccttcaattctgctgatcctgctgatcttctacggcatcgtgagcagcgttct gtataagaacagacgaacgaactgaaccacgctacctgcaatatctgatcgacttcaacagcgtgctgcagaccgacctggagaacattgttttcatcaaaat tgcgcgtgaaggaagcaaatctttaagagctggaaaaagcagcaacaacctggttaccgataaatagcagcagcctgaaggtggaaaacttcagcca gtttgttgtagcacctttaactatattagcctgagcctgatcctgattctgggcatctacttcaaaagcgtattttgaagttagcatcccgacctgattctgtacc aaaccagcgttagcctgatattagcttctttgaacagatcaagcaaacatcttcgagattgcgaaactgcaggtgtatcgggagaagcaaacgcacctgt tcaccaacaacgcgagcatcgaattccgatcgacaacgagcagagcagcgtgatccgttgcgataaagttaacttcagctacaacacacaaaagcgtgta tagcgggtatcaacctggaaattctgaaggcgagaaaaatggcatcctggcgcaagcggtagcggcaaaagcaccctgctgctgctgctgctgggtatg ctgctgtacgagcggcagctgacctatggcgtgagcgtatttaccagcgtatttggcgtgttctgcagaacatgacctggttccggacacatcctgaa gaacctggattgcgcgccgagcgacgaagatctggagcgtatttctgacacccgggtggcgaacgtgtgatcgcgaacacgtccgaacaagcgtgttag caaggttctgaaaagcggcaagaaactgagcgtggtggcagattcaaaaactgctgatcagccgtgacctgtagcagcgtgacatcattttctgggatg aagcgttttagcaacctggatgagaaaagcaagaaaagatctacaccgacgttctgagcaaccgttctatagcgataagagcgtggtgattgttagccac cagctggacatcgtgaactacgttgatagcatcattttgtggacaaaagcagcggcgatgttattaaggacacccacgataacctgatctaccgtaaccaa aactatcgtaaagctgttcggtctgaaagagaaggtgcacaacgacctgttccgcgtggcgagccatcatcaccatcatcaccacacccacataa

Supplementary Table 2 Primers used in this study (F, forward primer; R, reverse primer; Bold letters, mutated codon).

Primer name	Oligonucleotide sequence (5'-3')	Application
<i>sboA</i> pJL1 F	catcatcatcaccatattggcgaagcagcaaatgaac	Template construction for cell-free gene expression
<i>sboA</i> pJL1 R	tgtagcagccggctcgacttagctgcagcaggtaaacag	
<i>sboM</i> pET28a F	ccgcgcggcagccatattggtgaaatggagaacac	
<i>sboM</i> pET28a R	gtggtggtggtgctcgagttacttctcaaattgcgg	
<i>sboT</i> pET28a F	ccgcgcggcagccatattggtgcatgaaatcatt	
<i>sboT</i> pET28a R	gtggtggtggtgctcgagtttagtgggtggtggtg	
<i>sboA</i> pRSFDuet F	catcaccacagccaggatccgatggcgaagcagcaaatgaac	Plasmid construction for <i>in vivo</i> expression
<i>sboA</i> pRSFDuet R	cattatgcggccgaagcttttagctgcagcaggtaaacag	
<i>sboM</i> pRSFDuet F	catcaccacagccaggatccg atggcgaagcagcaaatgaac	
<i>sboM</i> pRSFDuet R	cattatgcggccgaagctt ttagctgcagcaggtaaacag	
<i>sboA</i> A-1R pRSFDuet F	gtcctgggtcgtggtggcggtgttatccagaccattagtggaggagctggacaacgtgctgggtcgtggtggcggt	
<i>sboA</i> A-1R pRSFDuet R	accgccaccacgaccagcacgtgtccagctcctcca	
<i>sboA</i> G-2R pRSFDuet F	acgtgctgggtcgtgcggtggcggtgttatccagacat	Site-specific mutagenesis
<i>sboA</i> G-2R pRSFDuet R	acaccgccaccgcacgaccagcacgtgtccagctcctc	
<i>sboA</i> Q6A F	ggtgttatc g taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6A R	gctaattggt ag cataacaccgccaccgcaccagcac	
<i>sboA</i> Q6C F	ggtgttatc t gaccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6C R	gctaattggt g cataacaccgccaccgcaccagcac	
<i>sboA</i> Q6D F	ggtgttatc g ataccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6D R	gctaattggt at cataacaccgccaccgcaccagcac	
<i>sboA</i> Q6E F	ggtgttatc g aaccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6E R	gctaattggt tt cataacaccgccaccgcaccagcac	
<i>sboA</i> Q6F F	ggtgttatc tt taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6F R	gctaattggt aa gataacaccgccaccgcaccagcac	
<i>sboA</i> Q6G F	ggtgttatc g gaccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6G R	gctaattggt g ccataacaccgccaccgcaccagcac	
<i>sboA</i> Q6H F	ggtgttatc ca taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6H R	gctaattggt at gataacaccgccaccgcaccagcac	
<i>sboA</i> Q6I F	ggtgttatc at taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6I R	gctaattggt at gataacaccgccaccgcaccagcac	
<i>sboA</i> Q6K F	ggtgttatc aa aaccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6K R	gctaattggt tt tgataacaccgccaccgcaccagcac	
<i>sboA</i> Q6L F	ggtgttatc ct taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6L R	gctaattggt ag gataacaccgccaccgcaccagcac	
<i>sboA</i> Q6M F	ggtgttatc at gaccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6M R	gctaattggt ca tataacaccgccaccgcaccagcac	
<i>sboA</i> Q6N F	ggtgttatc aa taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6N R	gctaattggt att gataacaccgccaccgcaccagcac	
<i>sboA</i> Q6P F	ggtgttatc ct taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6P R	gctaattggt ag gataacaccgccaccgcaccagcac	
<i>sboA</i> Q6R F	ggtgttatc cg taccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6R R	gctaattggt ac gataacaccgccaccgcaccagcac	
<i>sboA</i> Q6S F	ggtgttatc ag taccattagccacgaatgccgtatgaac	

<i>sboA</i> Q6S R	gctaattggt act gataaacaccgccacccgcaccagcac	
<i>sboA</i> Q6T F	ggtgttatc acg accattagccacgaatgccgtatgaac	
<i>sboA</i> Q6T R	gctaattggt ctg gataaacaccgccacccgcaccagcac	
<i>sboA</i> Q6V F	ggtgttatc gtt accattagccacgaatgccgtatgaac	
<i>sboA</i> Q6V R	gctaattggt aac gataaacaccgccacccgcaccagcac	
<i>sboA</i> Q6W F	ggtgttatc tgg accattagccacgaatgccgtatgaac	
<i>sboA</i> Q6W R	gctaattggt ccag ataaacaccgccacccgcaccagcac	
<i>sboA</i> Q6Y F	ggtgttatc tata ccattagccacgaatgccgtatgaac	
<i>sboA</i> Q6Y R	gctaattggt ataga ataaacaccgccacccgcaccagcac	
<i>sboA</i> L20 F	catcatcatcaccatattggcgaagcagcaaatgaac	
<i>sboA</i> L20A R	gtagcagccggctcgacttagctgcagcaggtaaa agc aattgccagct	
<i>sboA</i> L20C R	gtagcagccggctcgacttagctgcagcaggtaaa gcag aattgccagct	
<i>sboA</i> L20D R	gtagcagccggctcgacttagctgcagcaggtaaa atc aattgccagct	
<i>sboA</i> L20E R	gtagcagccggctcgacttagctgcagcaggtaaa ttc aattgccagct	
<i>sboA</i> L20F R	gtagcagccggctcgacttagctgcagcaggtaaa aa gaattgccagct	
<i>sboA</i> L20G R	gtagcagccggctcgacttagctgcagcaggtaaa gcc aattgccagct	
<i>sboA</i> L20H R	gtagcagccggctcgacttagctgcagcaggtaaa atg aattgccagct	
<i>sboA</i> L20I R	gtagcagccggctcgacttagctgcagcaggtaaa aat gaattgccagct	
<i>sboA</i> L20K R	gtagcagccggctcgacttagctgcagcaggtaaa ttf gaattgccagct	
<i>sboA</i> L20M R	gtagcagccggctcgacttagctgcagcaggtaaa cat gaattgccagct	
<i>sboA</i> L20N R	gtagcagccggctcgacttagctgcagcaggtaaa att gaattgccagct	
<i>sboA</i> L20P R	gtagcagccggctcgacttagctgcagcaggtaaa agg gaattgccagct	
<i>sboA</i> L20Q R	gtagcagccggctcgacttagctgcagcaggtaaa ctg gaattgccagct	
<i>sboA</i> L20R R	gtagcagccggctcgacttagctgcagcaggtaaa acg gaattgccagct	
<i>sboA</i> L20S R	gtagcagccggctcgacttagctgcagcaggtaaa act gaattgccagct	
<i>sboA</i> L20T R	gtagcagccggctcgacttagctgcagcaggtaaa agf gaattgccagct	
<i>sboA</i> L20V R	gtagcagccggctcgacttagctgcagcaggtaaa aac gaattgccagct	
<i>sboA</i> L20W R	gtagcagccggctcgacttagctgcagcaggtaaa ccag aattgccagct	
<i>sboA</i> L20Y R	gtagcagccggctcgacttagctgcagcaggtaaa ataga aattgccagct	
linearPF	cctacagcgtgagcattg	Construction of hybrid peptides
linearPR	cgcaccagcacgtgtc	
linearTF	gtcgaccggctgctaaca	
linearTR	cggattcagctgtaactca	
<i>lctA</i> _F1	aaaggcggcagtgaggattatcatacaatttctcatgaatgtaatatgaatagctggcaattt gtattacttctgctctctaa	
<i>lctA</i> _R1	ttaagagcagcaagtaataatacaaaattgccagctattcatattacattcatgagaaattgtatg aataactccactgccgccttt	
<i>lctA</i> _F	gacaacgtgctgggtgcgaaagggtgtagcgggtgtg	
<i>lctA</i> _R	tgtagcagccggctcgacttagctgcaacaagtga	
<i>rumA</i> _F1	ggtaacggcgttctgaagaccatctcccacgaatgcaacatgaacacctggcagttcctg ttcacctgctgctaa	
<i>rumA</i> _R1	ttagcagcaggtgaacaggaactgccaggtgttcatgttgcaattcgtgggagatgtcttc agaacgccgttacc	
<i>rumA</i> _F	gacaacgtgctgggtgcgggtaacggcgttctgaag	
<i>rumA</i> _R	tgtagcagccggctcgacttagcagcaggtgaacag	
<i>nuk</i> _F1	aaaaaaaaatctggcgttatcccagccgtttctcacgattgccacatgaacagcttccagtt cgtgttcacctgctgctctaa	
<i>nuk</i> _R1	ttaggagcagcaggtgaacacgaactggaagctgttcatgtggcaatcgtgagaaacgg tcgggataacgccagattttttt	
<i>nuk</i> _F	gacaacgtgctgggtgcg aaaaaaaaaatctggcgttctc	
<i>nuk</i> _R	tgtagcagccggctcgac ttaggagcagcaggtgaa	
<i>mutII</i> _F1	aatcgctggtggcagggcgtagttccgactgtatcttacgaatgtcgtatgaactcttgcca gcacgtgttcaactgctgctaa	

<i>mutII</i> _R1	ttagcagcaagtgaacacgtgctgccaaggttcatacagacattcgtaagatacagtcgg aactacgccctgccaccagcgatt
<i>mutII</i> _F	gacaacgtgctgggtgcg aatcgctgggtggcaggg
<i>mutII</i> _R	tgtagcagccggtcgac ttagcagcaagtgaaca
<i>salA</i> _F1	aaacgtggtagcgggttgattgctaccatcaccgatgattgccgaactccgtattgtttgc tgctaa
<i>salA</i> _R1	ttagcagcaaaacaatacggagttcggacaatcatcggtgatgtagcaatccaaccgct accacgttt
<i>salA</i> _F	gacaacgtgctgggtgcg aaacgtggtagcgggtg
<i>salA</i> _R	tgtagcagccggtcgac ttagcagcaaaacaatac
Lan1_F1	ggtaacggcgtgtgaccaccatctctcacgaatgcaacatgaacagctggcagttcctg ttcactgttgctaa
Lan1_R1	ttagcaacaggtgaacaggaactgccagctgttcatgttgattcgtgagagatgggtgctc acaacgccgttacc
Lan1_F	gacaacgtgctgggtgcggtgaacggcgttgacc
Lan1_R	tgtagcagccggtcgac ttagcaacaggtgaacag
Lan2_F1	ggcaacggcgttatcaaaacatctcccacgaatgcgctatgaacacctggcagttcattt tcactgctgctctaa
Lan2_R1	ttaggagcagcaggtgaaaatgaactgccaggtgttcatagcgcattcgtgggagatggt tttgataacgccgttgcc
Lan2_F	gacaacgtgctgggtgcggtgaacggcgttatcaaa
Lan2_R	tgtagcagccggtcgacttaggagcagcaggtgaa
Lan3_F1	ggcgatggtgttatgcataccctgacctgaatgtcatgaacactggcagttcctgct gacctgttgctaa
Lan3_R1	ttagcaacaggtcagcaggaactgccaagtgttcatgtgacattcatgggtcagggatg cataacaccatcgcc
Lan3_F	gacaacgtgctgggtgcggtgatggtgttatgcat
Lan3_R	tgtagcagccggtcgacttagcaacaggtcagcag
Lan4_F1	aacggtgttatcaccacgattagccacgaatgccacctgaacacctgggccttcgcgttc acctgttgctctaa
Lan4_R1	ttaagagcaacaggtgaacgcgaaggcccaggtgttcaggtggcattcgtggctaactg tggtgataaacacgtt
Lan4_F	gacaacgtgctgggtgccaacggtgttatcaccacg
Lan4_R	tgtagcagccggtcgacttaagagcaacaggtgaa
Lan5_F1	ggtaacggtgtttccacactatttctcacgaatgttactacaactcctggccttcatttcac ctgctgctaa
Lan5_R1	ttagcagcaggtgaaaatgaaagcccaggagttgtagtaacattcgtgagaaatagtgtg gaaaacaccgttacc
Lan5_F	gacaacgtgctgggtgcggtgaacggtgtttccac
Lan5_R	tgtagcagccggtcgacttagcagcaggtgaaaat
Lan6_F1	ggtaacggcgtatccgtaccatcagccatgaatgtcacatgaacacctggcagttcctgt ttacctgttgctctaa
Lan6_R1	ttaagagcaacaggtaaacaggaactgccaggtgttcatgtgacattcatggctgatgta cggatagcgccgttacc
Lan6_F	gacaacgtgctgggtgcggtgaacggcgtatccgt
Lan6_R	tgtagcagccggtcgacttaagagcaacaggtaaa
Lan7_F1	ggcaacggtcagtttaacactatttctcatgaatgccactggaacacctggcagttcatgtt cacctgctgttctaa
Lan7_R1	ttaagaacagcaggtgaacatgaactgccaggtgttccagtggtcattcatgagaaatagt gttaaaactgaccgttgcc
Lan7_F	gacaacgtgctgggtgcggtgaacggtcagtttaac
Lan7_R	tgtagcagccggtcgacttaagaacagcaggtgaa
Lan8_F1	cgcaaactgtgtgcatcaacaccgtttccacgaatgcaacatgaactcttggcagttcgt gttcacctgctgcgctaa
Lan8_R1	ttacgcgacagcaggtgaacacgaactgccaaggttcatgttgattcgtgggaaacgggt gttgatgacaccagattgctg
Lan8_F	gacaacgtgctgggtgcg cgcaaactgtgtgcatc

Lan8_R	tgtagcagccggctgacttacgcgcagcaggtgaa	
Lan9_F1	gataaaggtcagttcgacaccatctctcacgaatgtcattggaacacctggcagtttctgttt acctgttgctcctaa	
Lan9_R1	ttaggagcaacaggtaaacagaaactgccaggtgttccaatgacattcgtgagagatggt gtcgaactgacctttatc	
Lan9_F	gacaacgtgctgggtgctgataaaggctcagttcgac	
Lan9_R	tgtagcagccggctgacttaggagcaacaggtaaa	
Lan10_F1	ggtaacggtgtaatacaaaactattagccatgaatgccacatgaactcctggcagtggttt cacctgctgctcttaa	
Lan10_R1	ttaagagcagcaggtgaaaaatccactgccaggagttcatgtggcattcatggctaatagtt ttgattacaccgttacc	
Lan10_F	gacaacgtgctgggtgctgggtaacggtgtaatacaaa	
Lan10_R	tgtagcagccggctgacttaagagcagcaggtgaa	
Lan11_F1	gggtctggcgtaatcaagactatcagccacgaatgccacatgaactcttgccagggcgatc ttcacttgctgctcctaa	
Lan11_R1	ttaggagcagcaagtgaagatcgctgccaaggtcatgtggcattcgtggctgatatgt cttgattacgccagaacc	
Lan11_F	gacaacgtgctgggtgctgggttctggcgtaatcaag	
Lan11_R	tgtagcagccggctgacttaggagcagcaagtga	
Lan12_F1	ggtaacggtgtgtttataacctgactcatgaatgtaacctggcgacctggacgaaaaaac tgaaatgctgctaa	
Lan12_R1	ttagcagcatttcagtttttctccaggtcgccaggttacattcatgagtcagggtataaac aacaccgttacc	
Lan12_F	gacaacgtgctgggtgctgggtaacggtgtgtttat	
Lan12_R	tgtagcagccggctgacttagcagcatttcagttt	
<i>degP</i> H1F	ttgcagcattacacgtcttggggctattagcggattattag	Construction of homologous fragments for gene deletion
<i>degP</i> H1R	tccagcctacacaatcgctcctcgattaacagataacgca	
<i>degP</i> H2F	gatattcatatggaccatgggggttctccttacaatctgtg	
<i>degP</i> H2R	gacatgggaattagccatgcctgcgccaacaagtgcata	
<i>ydgD</i> H1F	ttgcagcattacacgtcttggcagggcgctaaaaagcatcat	
<i>ydgD</i> H1R	tccagcctacacaatcgctccgcatatcacactctggtgagta	
<i>ydgD</i> H2F	gatattcatatggaccatggccggttttcgcgacaagctggatc	
<i>ydgD</i> H2R	gacatgggaattagccatgcgctaaacctgaactggaggtga	
<i>ypdE</i> H1F	ttgcagcattacacgtcttgccttgccagcacattgccgtacagt	
<i>ypdE</i> H1R	tccagcctacacaatcgctcgcccttattcaacgtcttacgcgtga	
<i>ypdE</i> H2F	gatattcatatggaccatgggatctgccgcacttctgttccgag	
<i>ypdE</i> H2R	gacatgggaattagccatgcctgcaggcacagctgcgacaatc	
<i>degQ</i> H1F	ttgcagcattacacgtcttgcgcatcattattggtgctgtg	
<i>degQ</i> H1R	tccagcctacacaatcgctccggctaacaatgagacaccagatta	
<i>degQ</i> H2F	gatattcatatggaccatggccggcatcaggcttacgtgtgat	
<i>degQ</i> H2R	gacatgggaattagccatgccaccagcaatgcttcaatacacga	
<i>hslV</i> H1F	ttgcagcattacacgtcttgggtggatcagcacgtcgagaatacgt	
<i>hslV</i> H1R	tccagcctacacaatcgctcgcgacatttgcatctataccaacca	
<i>hslV</i> H2F	gatattcatatggaccatggggcgtacgttactatagtgtcac	
<i>hslV</i> H2R	gacatgggaattagccatgcggttctctcagcagccgtacca	
<i>pepN</i> H1F	ttgcagcattacacgtcttggggcgaaaagccactgatat	
<i>pepN</i> H1R	tccagcctacacaatcgctccagcttattctgttgcgta	
<i>pepN</i> H2F	gatattcatatggaccatggggcgtggaaaatctcttgccgat	
<i>pepN</i> H2R	gacatgggaattagccatgcgaccgggattattgttcttga	
pkD4 attP/attB F	agaaggccatcctgacggat	Deletion cassette amplification
pkD4 attP/attB R	gcccttagagcctctcaaag	
<i>degP</i> F	gggtcattagcggattattag	
<i>degP</i> R	cctgcgccaacaagtgcata	
<i>ydgD</i> F	gcaggcggctaaaaagcatcat	

<i>ydgD</i> R	cgcgtaaacctgaactggaggtga	
<i>ypdE</i> F	cgttgccagcacattgccgtacagt	
<i>ypdE</i> R	cctgcaggcacagctcgagcaatc	
<i>pepN</i> F	gggcagaaagccactgatat	
<i>pepN</i> R	cgaccgggattattgtgcttga	
<i>degQ</i> F	cggcatcattattgtgctgtg	
<i>degQ</i> R	cccaccagcaatgcttcaaatacacga	
<i>hslV</i> F	gtgggatcagcacgtcgagaatacgt	
<i>hslV</i> R	cggcttcttcagcagccgtacca	
<i>kan</i> MR	cctcgtcctgcagttcattca	Deletion verification
<i>degP</i> veriF	gccatccagatgtcgagcagct	
<i>pepN</i> veriF	ctcgcgtaaccagttaagatagt	
<i>ydgD</i> veriF	cagcataaagcagcacctg	
<i>ypdE</i> veriF	aggctgtcgctttgtaacagggtca	
<i>degQ</i> veriF	gggttagtcgtcggcatcat	
<i>hslV</i> veriF	gacgggtcctgctgctgttcggt	

Supplementary Table 3 Plasmids constructed in this study.

Plasmid	Antibiotic	Application
pJL1 <i>sboA</i>	50 µg/mL kanamycin	Templates used for cell-free gene expression
pET28a- <i>sboM</i>		
pET28a- <i>sboT</i>		
pJL1- <i>sboA</i> Q6/L20 (a total of 53 mutated plasmids)		Plasmids used for <i>in vivo</i> expression
pRSFDuet-1- <i>sboA-sboM</i>		
pRSFDuet-1- <i>sboA</i> A-1R- <i>sboM</i>		
pRSFDuet-1- <i>sboA</i> G-2R- <i>sboM</i>		
pRSFDuet-1- <i>sboA</i> A-1R Q6H- <i>sboM</i>		
pRSFDuet-1- <i>sboA</i> A-1R L20I- <i>sboM</i>		
pRSFDuet-1- <i>sboA</i> A-1R L20M- <i>sboM</i>		
pRSFDuet-1- <i>sboA</i> A-1R L20I Q6H- <i>sboM</i>		
pRSFDuet-1- <i>sboA</i> A-1R L20M Q6H- <i>sboM</i>		
pRSFDuet-1- <i>sboA</i> A-1R L20M Q6K- <i>sboM</i>		
pKD4 <i>attP/attB-degP</i>	50 µg/mL kanamycin 100 µg/mL ampicillin	Plasmids used for protease/peptidase deletion
pKD4 <i>attP/attB-degQ</i>		
pKD4 <i>attP/attB-hslV</i>		
pKD4 <i>attP/attB-pepN</i>		
pKD4 <i>attP/attB-ydgD</i>		
pKD4 <i>attP/attB-ypdE</i>		
pKD46	100 µg/mL ampicillin	Expression of λ-Red recombinases
pYF2	100 µg/mL ampicillin	Expression of TP901-1 integrase

Supplementary Table 4 Proteases and peptidases deleted in the genome of *E. coli* BL21 Star (DE3).

Gene	Protein product	Activity	Reference
<i>degP</i>	Serine endoprotease, DegP	DegP and DegQ may degrade transiently denatured and unfolded proteins, which accumulate in the periplasm (Cleavage sites: Val/Xaa or Ile/Xaa).	1, 2
<i>degQ</i>	Serine endoprotease, DegQ		1, 2
<i>ydgD</i>	Serine endoprotease, YdgD	-	UniProt: P76176
<i>hslV</i>	ATP-dependent protease subunit, HslV	Protease subunit of a proteasome-like degradation complex believed to be a general protein degrading machinery.	3
<i>pepN</i>	Aminopeptidase N, PepN	PepN is responsible for the majority of the aminopeptidase activity in <i>E. coli</i> and that it prefers to cleave basic and small amino acids at the amino terminus of substrates.	4
<i>ypdE</i>	Aminopeptidase, YpdE	YpdE has a broad aminopeptidase activity on non-blocked peptides by progressively cleaving amino acids off the peptide substrate. Aminopeptidase activity stops at the residue before the first proline in the peptide.	5

Supplementary Table 5 The yields of salivaricin B variants synthesized in UniBioCat reactions.

Variant	MW (Da) ^a	Yield (μM)	Variant	MW (Da) ^a	Yield (μM)
SalB_Q6A	2747	115	SalB_L20F-Q6A	2781	33
SalB_Q6C	2779	19	SalB_L20F-Q6H	2847	51
SalB_Q6E	2805	12	SalB_L20F-Q6K	2838	40
SalB_Q6F	2823	5	SalB_L20F-Q6M	2841	28
SalB_Q6G	2733	21	SalB_L20F-Q6N	2824	25
SalB_Q6H	2813	26	SalB_L20I-Q6A	2765	35
SalB_Q6I	2789	13	SalB_L20I-Q6H	2831	31
SalB_Q6K	2804	32	SalB_L20I-Q6K	2822	30
SalB_Q6L	2789	23	SalB_L20I-Q6M	2825	48
SalB_Q6M	2807	23	SalB_L20I-Q6N	2825	84
SalB_Q6N	2790	50	SalB_L20M-Q6A	2747	55
SalB_Q6S	2763	44	SalB_L20M-Q6H	2813	54
SalB_Q6V	2775	23	SalB_L20M-Q6K	2804	33
SalB_Q6W	2862	9	SalB_L20M-Q6M	2807	31
SalB_Q6Y	2839	25	SalB_L20M-Q6N	2790	59
SalB_L20A	2762	30			
SalB_L20C	2794	15			
SalB_L20E	2820	22			
SalB_L20F	2838	33			
SalB_L20H	2828	39			
SalB_L20I	2804	71			
SalB_L20K	2819	21			
SalB_L20M	2822	65			
SalB_L20N	2805	93			
SalB_L20Q	2819	12			
SalB_L20R	2847	16			
SalB_L20S	2778	15			
SalB_L20V	2790	44			
SalB_L20W	2877	52			
SalB_L20Y	2854	14			

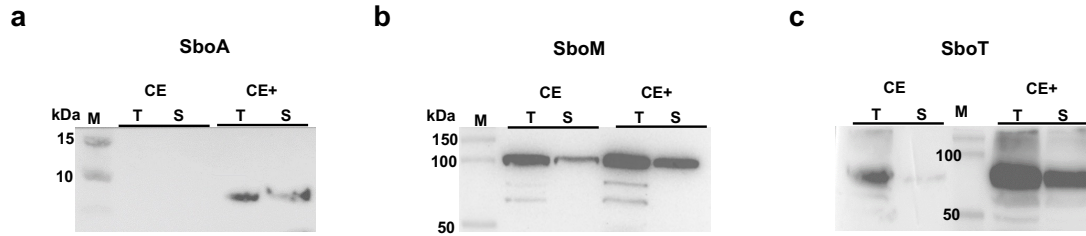
^a MW, molecular weight.

Supplementary Table 6 Minimal inhibitory concentrations (MIC) and IC₅₀ of salivaricin B and variants.

Peptides	<i>Bacillus subtilis</i>		<i>Micrococcus luteus</i>		<i>Lactococcus lactis</i>	
	MIC (μ M)	IC ₅₀ (μ M)	MIC (μ M)	IC ₅₀ (μ M)	MIC (μ M)	IC ₅₀ (μ M)
Salivaricin B	4	1.88±0.01	2	1.19±0.05	4	1.67±0.08
SalB_Q6H	1	0.52±0.03	0.5	0.34±0.05	1	0.58±0.10
SalB_L20M	4	1.82±0.05	1	0.58±0.02	4	1.97±0.03
SalB_L20I-Q6H	2	1.56±0.30	2	0.84±0.17	2	1.07±0.13
SalB_L20M-Q6H	4	1.83±0.12	2	1.19±0.10	4	1.71±0.17

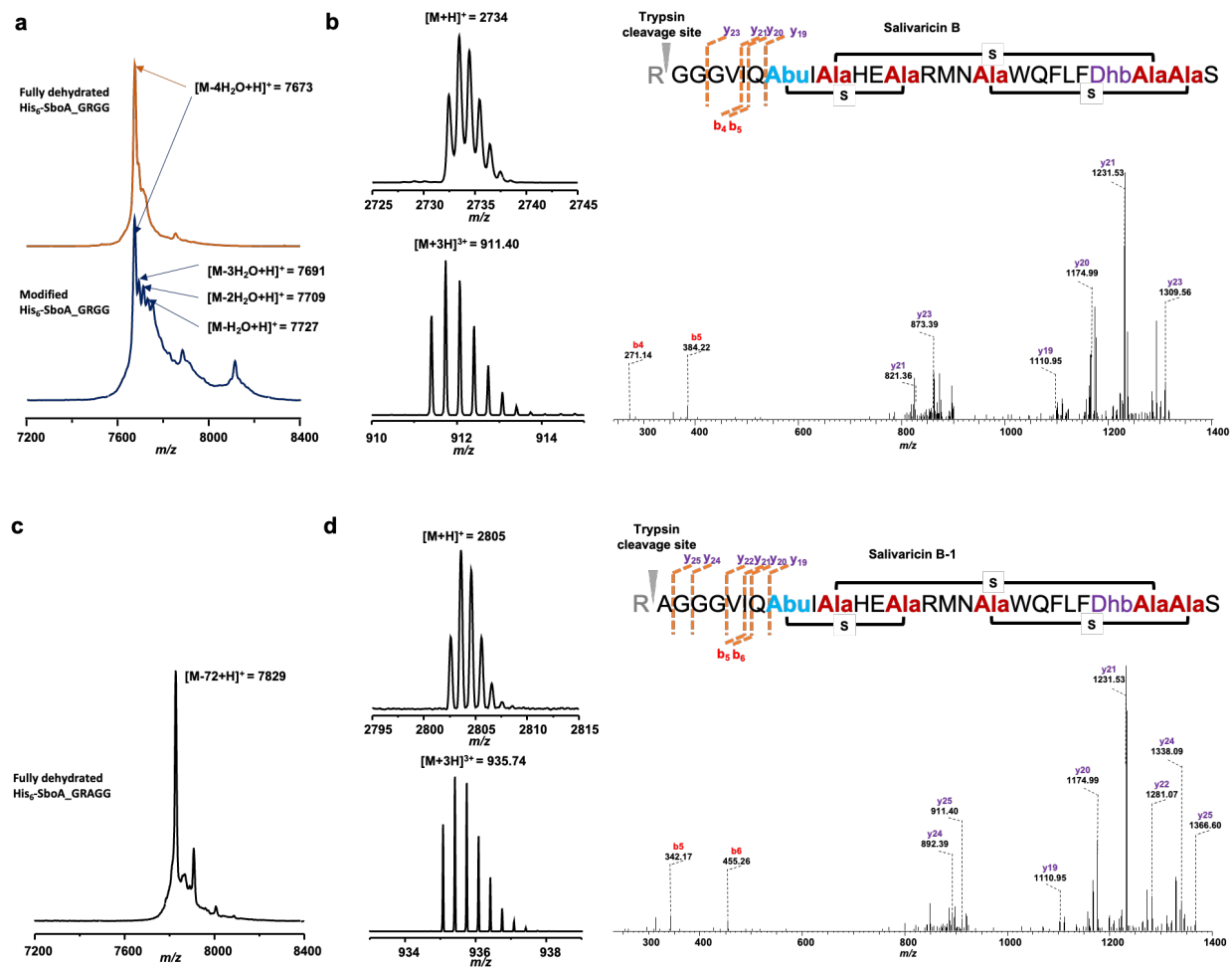
Source data are provided as a Source Data file.

Supplementary Figures



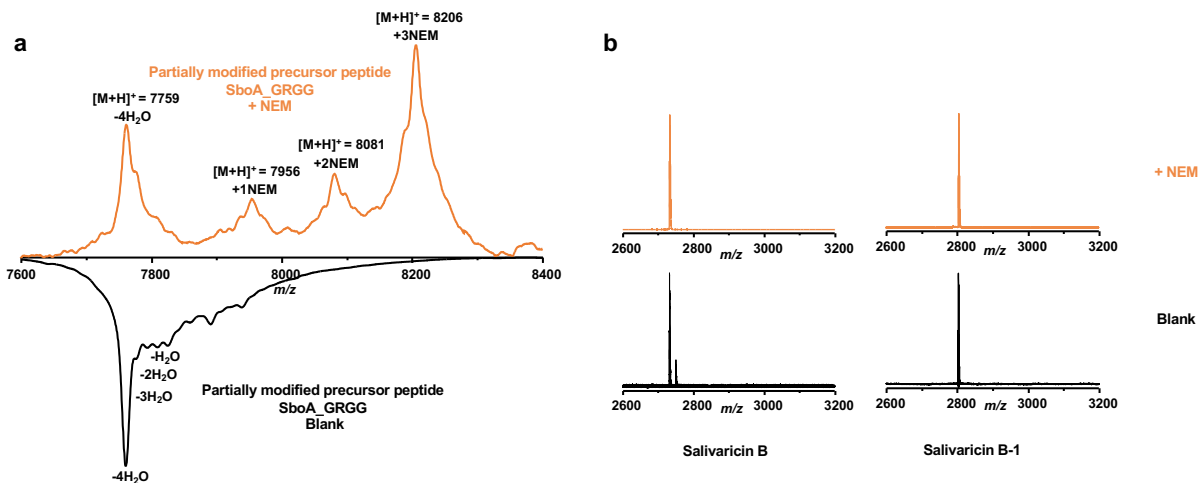
Supplementary Figure 1. Cell-free expression of salivarin B biosynthesis genes.

Western-blot analysis of cell-free expressed (a) SboA (7.8 kDa), (b) SboM (113 kDa), and (c) SboT (83 kDa). Peptide/enzymes are labeled with the anti-His antibody. CE, cell extracts prepared from *E. coli* BL21 Star (DE3); CE+, CE enriched with chaperones (DanK-DnaJ-GrpE and GroES-GroEL); M, protein marker; T, total protein; S, soluble protein. Results were reproduced three times independently; representative data are shown. Source data are provided as a Source Data file.



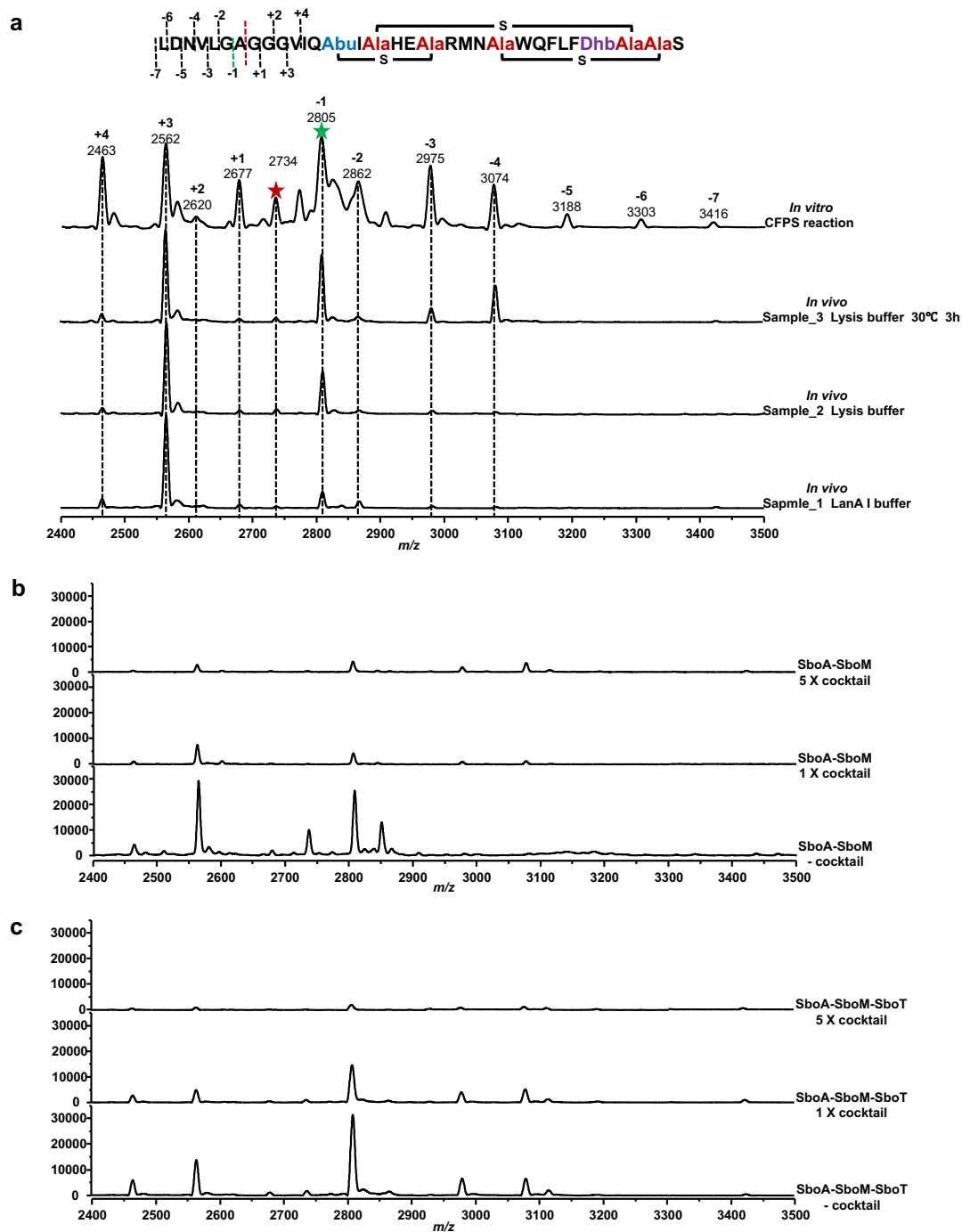
Supplementary Figure 2. *In vivo* biosynthesis and identification of salivarin B and salivarin B-1.

(a) MALDI-TOF-MS analysis of modified precursor peptide of SboA_A-1R. (b) MALDI-TOF-MS (top left), LC-MS (bottom left), and MS/MS (right) analysis of matured salivarin B (*in vitro* maturation by trypsin digestion). (c) MALDI-TOF-MS analysis of fully dehydrated precursor peptide of SboA_G-2R. (d) MALDI-TOF-MS (top left), LC-MS (bottom left), and MS/MS (right) analysis of matured salivarin B-1 (*in vitro* maturation by trypsin digestion).



Supplementary Figure 3. Identification of the thioether formation by using the *N*-ethylmaleimide (NEM) derivatization method.

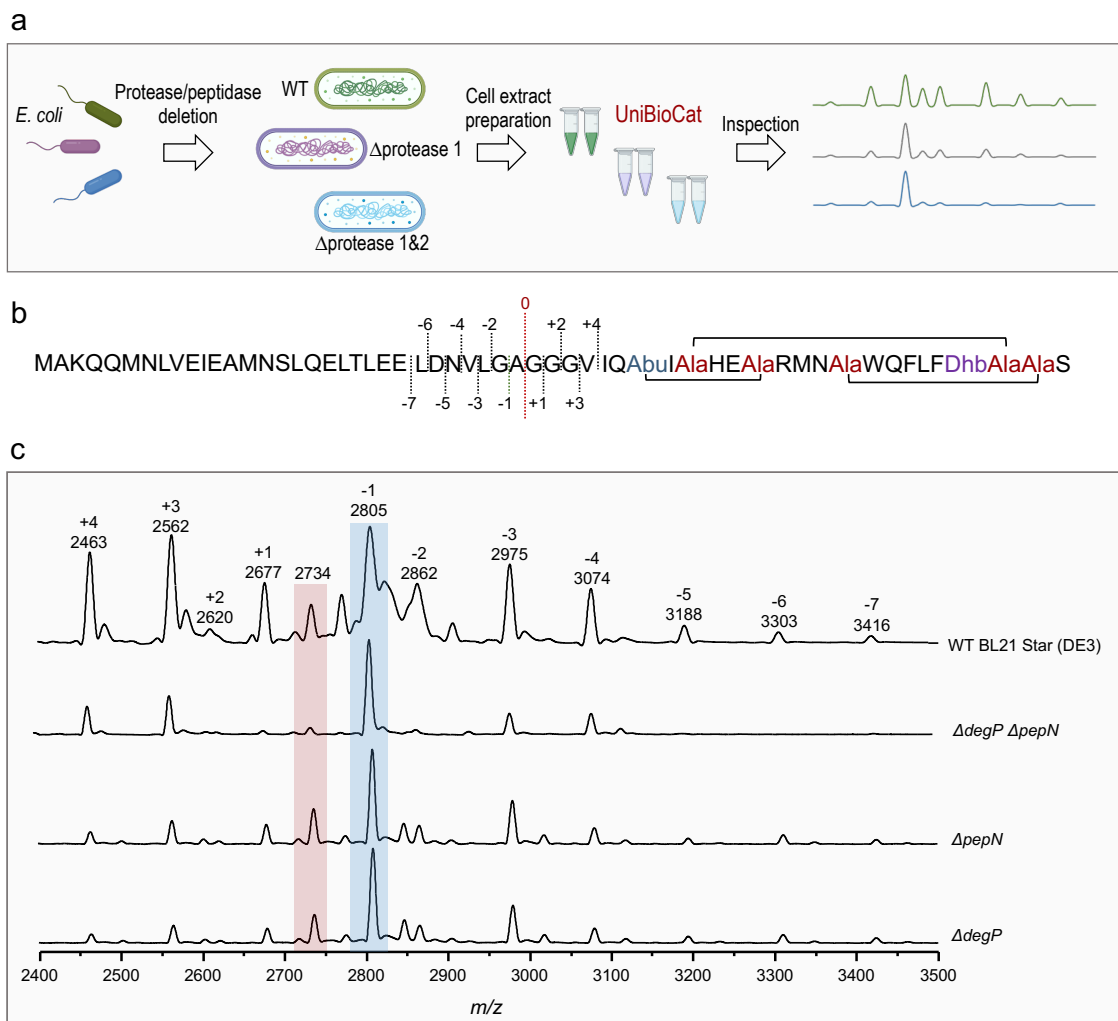
(a) MALDI-TOF-MS (linear positive mode) identification of partially modified precursor peptide SboA_A-1R treated with NEM. Note that the masses indicated as +1NEM, +2NEM, and +3NEM are non-dehydrated peptide peaks. (b) MALDI-TOF-MS (reflectron positive mode) identification of mature salivarin B and salivarin B-1. In the alkylation assay, NEM is added to the thiol (-SH) group of the cysteine residue and the derivatives show a molecular mass increase of 125 Da per reaction. The partially modified SboA_A-1R was a peptide mixture purified from the *in vivo* co-expression of sboA_A-1R and sboM for 1 h at 30°C, in which serine and threonine residues were not completely dehydrated to form the thioether crosslinks leaving 1-3 cysteine residues to react with NEM. When NEM was added, the peptide molecular mass increased accordingly. For the fully modified peptide, no NEM was added and thus the peptide molecular mass was not changed as observed in (b) for both salivarin B and salivarin B-1. The consistent molecular weights indicate the successful modification of the thioether crosslinks in the mature peptides.



Supplementary Figure 4. Effects of endogenous potential proteases on salivaricin B synthesis.

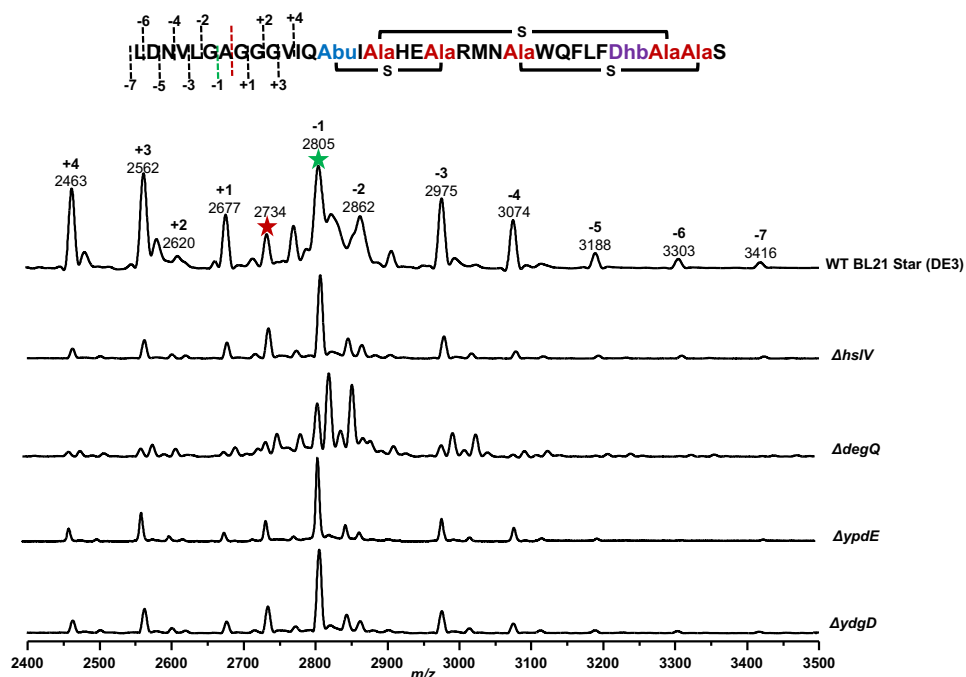
(a) Generation of salivaricin B analogs by non-specific protease degradation both *in vivo* and *in vitro*. The red and green stars indicate salivaricin B and salivaricin B-1, respectively. Analogs detected *in vitro* were products from cell-free co-expression of *sboA*, *sboM*, and *sboT*. Analogs detected *in vivo* were products from co-expression of *sboA* and *sboM*. Harvested strains were divided into three parts undergoing different treatments. Sample 1 was suspended in LanA I buffer containing 4 M guanidine hydrochloride and then sonicated for cell lysis, in which the endogenous proteases were inactivated by guanidine hydrochloride. Sample 2 was suspended

in lysis buffer, sonicated, and then immediately placed on ice. Sample 3 was suspended in lysis buffer and incubated at 30°C for 3 h after sonication, in which the endogenous proteases conducted catalysis. All of samples were then centrifuged and the supernatants were desalted by C18 ZipTip and analyzed by MALDI-TOF-MS with linear positive mode. **(b)** Products from cell-free reactions (SboA and SboM) with increased concentrations of protein inhibitor cocktail in the UniBioCat reactions. **(c)** Products from cell-free reactions (SboA, SboM, and SboT) with increased concentrations of protein inhibitor cocktail in the UniBioCat reactions. The catalytic activities of endogenous proteases/peptidases and SboT were inhibited by a high concentration of protein inhibitor (5x).



Supplementary Figure 5 Strain engineering to reduce non-specific protease/peptidase degradation in UniBioCat reactions.

(a) Schematic diagram of protease/peptidase deletion for the preparation of enhanced cell extracts for UniBioCat reactions. Created with BioRender.com. (b) Observed products by non-specific degradation with endogenous proteases and/or peptidases. (c) MALDI-TOF-MS analysis of salivaricin B (highlighted in red), salivaricin B-1 (highlighted in blue), and other non-specifically degraded products. Cell extracts used for UniBioCat were prepared from the wild-type (WT) strain *E. coli* BL21 Star (DE3) and three other engineered strains with protease (*degP*) and/or peptidase (*pepN*) gene(s) deleted. Note that SboM was pre-expressed in each engineered strain for the preparation of SboM-enriched cell extracts, which were then used for cell-free co-expression of SboA and SboT.

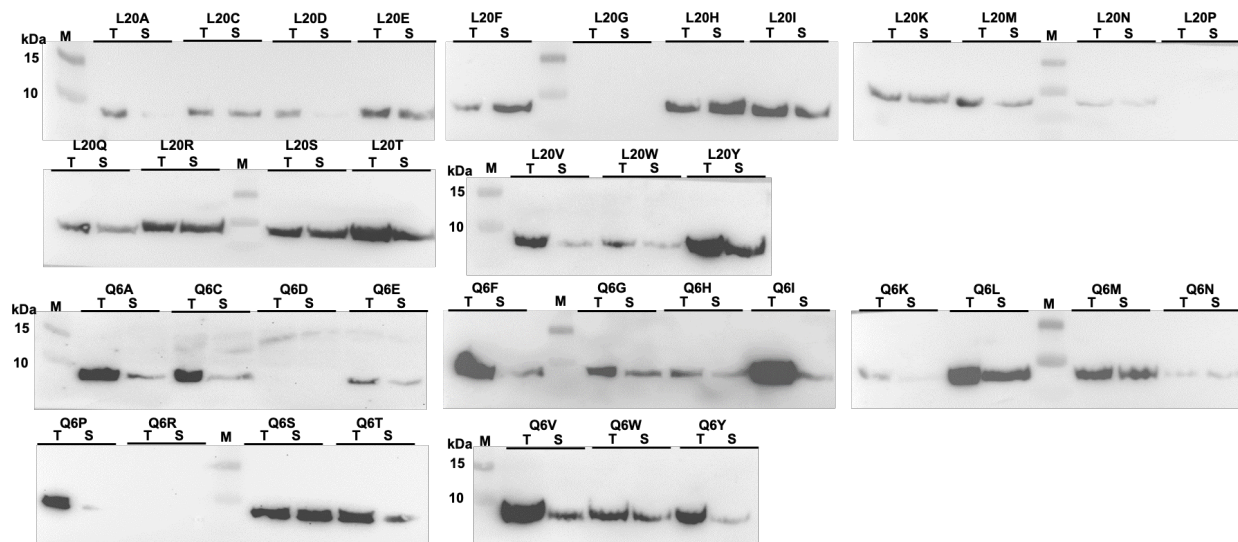


Supplementary Figure 6. Reduction of the non-specific protease/peptidase degradation in UniBioCat reactions.

Top, observed products by non-specific degradation with endogenous proteases/peptidases. Bottom, MALDI-TOF-MS analysis of salivarin B (red star), salivarin B-1 (blue star), and other non-specifically degraded products. Cell extracts used for UniBioCat were prepared from the wild-type (WT) strain *E. coli* BL21 Star (DE3) and four engineered strains with protease or peptidase deleted.

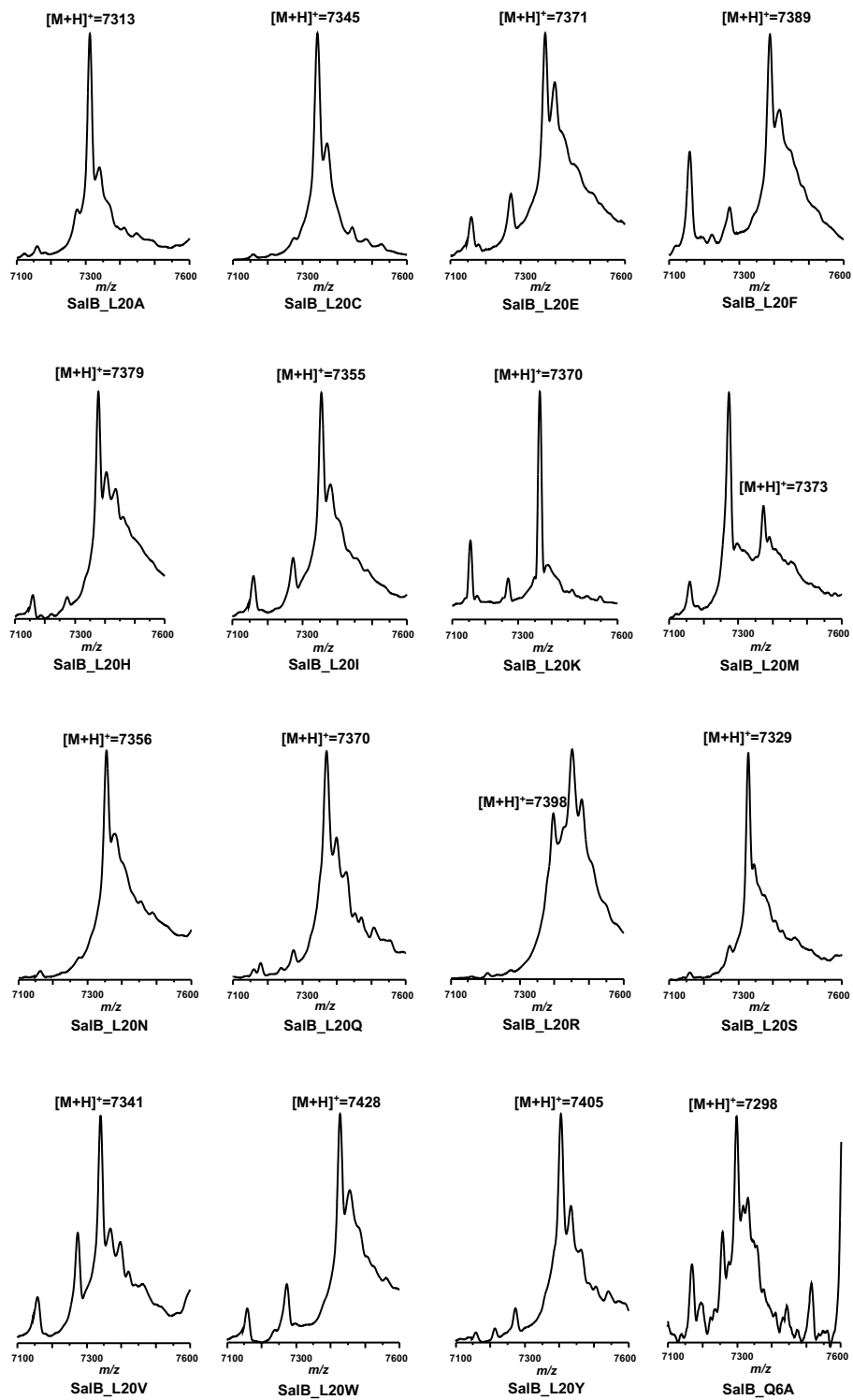
	Core peptides																																																											
	Q6																		L20																																									
Salivaricin B	M	A	K	Q	Q	M	N	L	V	-	-	E	I	E	A	M	N	S	L	Q	E	L	T	L	E	E	L	D	N	V	L	G	A	-	-	G	G	G	V	I	Q	T	I	S	H	E	C	R	M	N	S	W	Q	F	L	F	T	C	C	S
Lacticin 481	-	-	-	-	-	-	M	K	-	-	-	E	Q	N	S	F	N	L	L	Q	E	V	T	E	S	E	L	D	L	I	L	G	A	K	G	G	S	G	V	I	H	T	I	S	H	E	C	N	M	N	S	W	Q	F	V	F	T	C	C	S
RKV69777	-	-	-	-	-	M	N	L	V	-	-	E	I	E	A	M	N	S	L	Q	E	L	T	L	E	E	L	D	N	V	L	G	A	-	-	G	D	G	V	I	Q	T	I	S	H	E	C	R	M	N	S	W	Q	F	L	F	T	C	C	S
WP_002175376	-	-	-	-	-	M	K	Q	V	-	-	E	L	E	A	L	Q	A	I	N	D	L	T	D	E	E	L	E	E	I	V	G	A	-	-	G	S	G	V	I	K	T	I	S	H	E	C	H	M	N	S	W	Q	A	I	F	T	C	C	S
WP_009402788	-	-	-	-	-	M	S	N	I	-	-	N	M	Q	A	A	A	L	D	E	L	S	D	A	E	L	D	Q	V	L	G	A	-	-	N	G	V	V	I	Y	T	L	S	H	E	C	H	M	N	S	W	Q	F	L	F	T	C	C	S	
WP_145517223	-	-	-	-	-	M	N	L	V	-	-	E	I	E	A	M	N	S	L	Q	E	L	T	L	E	E	L	D	N	V	L	G	A	-	-	G	G	G	V	I	Q	T	I	S	H	E	C	R	M	N	S	W	Q	F	L	F	T	C	C	S
WP_121564619	-	-	-	-	-	M	K	D	T	N	I	D	I	E	A	T	N	T	L	Q	E	L	S	L	D	E	E	L	D	T	I	G	A	G	K	-	E	G	V	P	T	I	S	H	D	C	H	M	N	S	W	Q	F	L	F	T	C	C	S	
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WP_041790396	-	-	-	-	-	M	K	K	-	-	-	D	I	E	L	M	N	T	I	Q	E	V	S	L	E	E	L	D	Q	I	G	A	G	K	-	N	G	V	F	K	T	I	S	H	E	C	H	M	N	S	W	Q	F	L	F	T	C	C	S	
WP_002935048	-	-	-	-	-	M	K	K	-	-	-	D	I	E	L	M	S	T	I	Q	E	V	S	L	E	E	L	D	Q	I	G	A	G	K	-	N	G	V	F	K	T	I	S	H	E	C	H	M	N	S	W	Q	F	L	F	T	C	C	S	
WP_0037580734	-	-	-	-	-	M	K	D	-	-	-	N	Y	E	L	M	N	T	I	Q	E	V	S	L	E	E	L	D	Q	I	G	A	G	K	-	N	G	A	I	K	T	I	S	H	E	C	H	M	N	S	W	Q	F	L	F	T	C	C	S	
WP_002935049	-	-	-	-	-	M	K	D	-	-	-	N	Y	E	L	M	N	T	I	Q	E	V	S	L	E	E	L	D	Q	I	G	A	G	K	-	N	G	V	F	K	T	I	S	H	E	C	H	M	N	S	W	Q	F	L	F	T	C	C	S	
WP_105257777	-	-	-	-	-	-	M	E	L	M	S	T	I	Q	E	V	S	L	E	E	L	E	L	D	Q	I	G	A	G	K	-	N	G	V	F	K	T	I	S	H	E	C	H	M	N	S	W	Q	F	L	F	T	C	C	S					

Supplementary Figure 7. Alignment of homologous peptides based on the salivarin B precursor peptide. Amino acid sequences of the homologous peptides were aligned by using MEGA. Q6 and L20 were selected for the mutation.

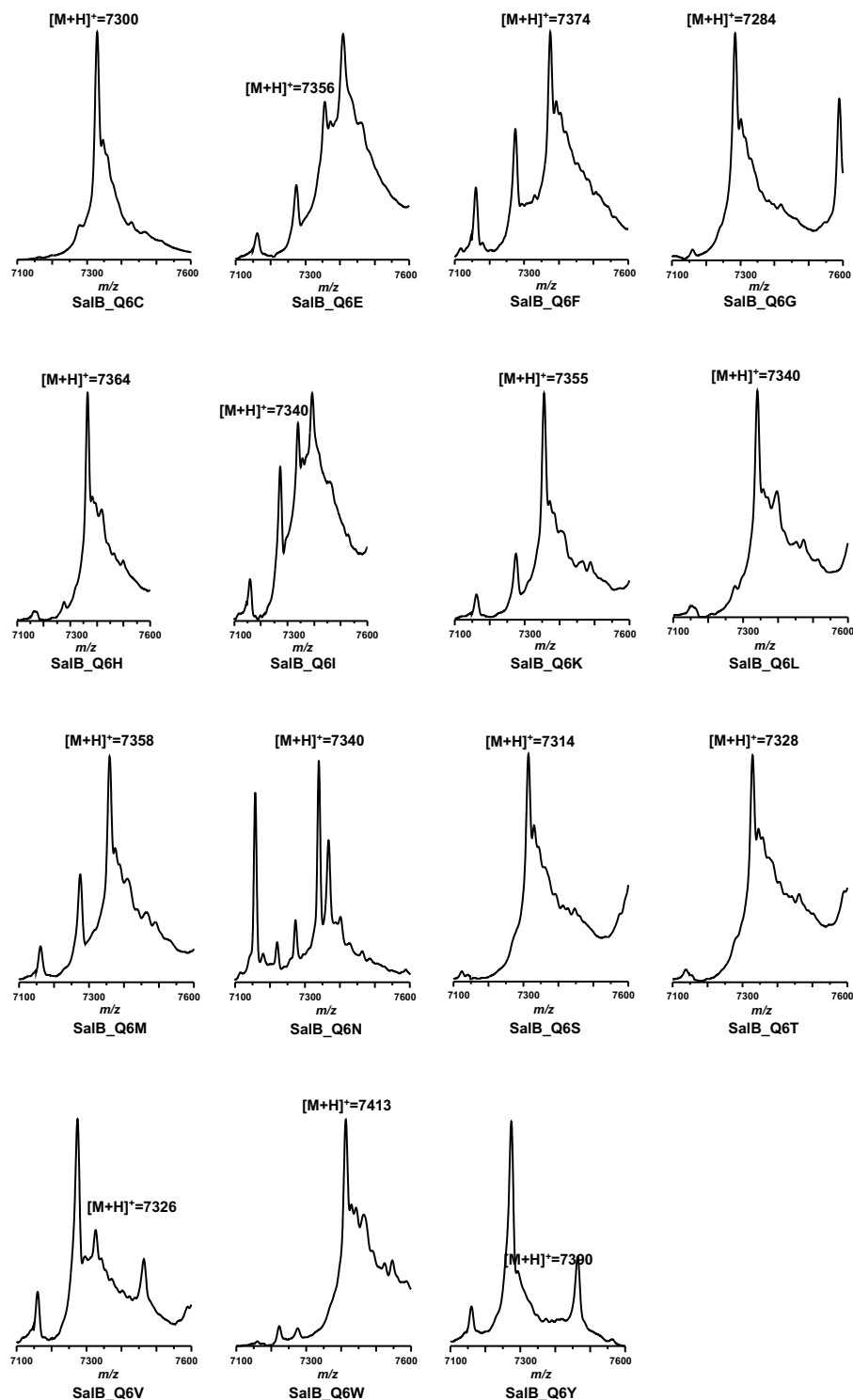


Supplementary Figure 8. Western-blot analysis of cell-free expressed mutated precursor peptides.

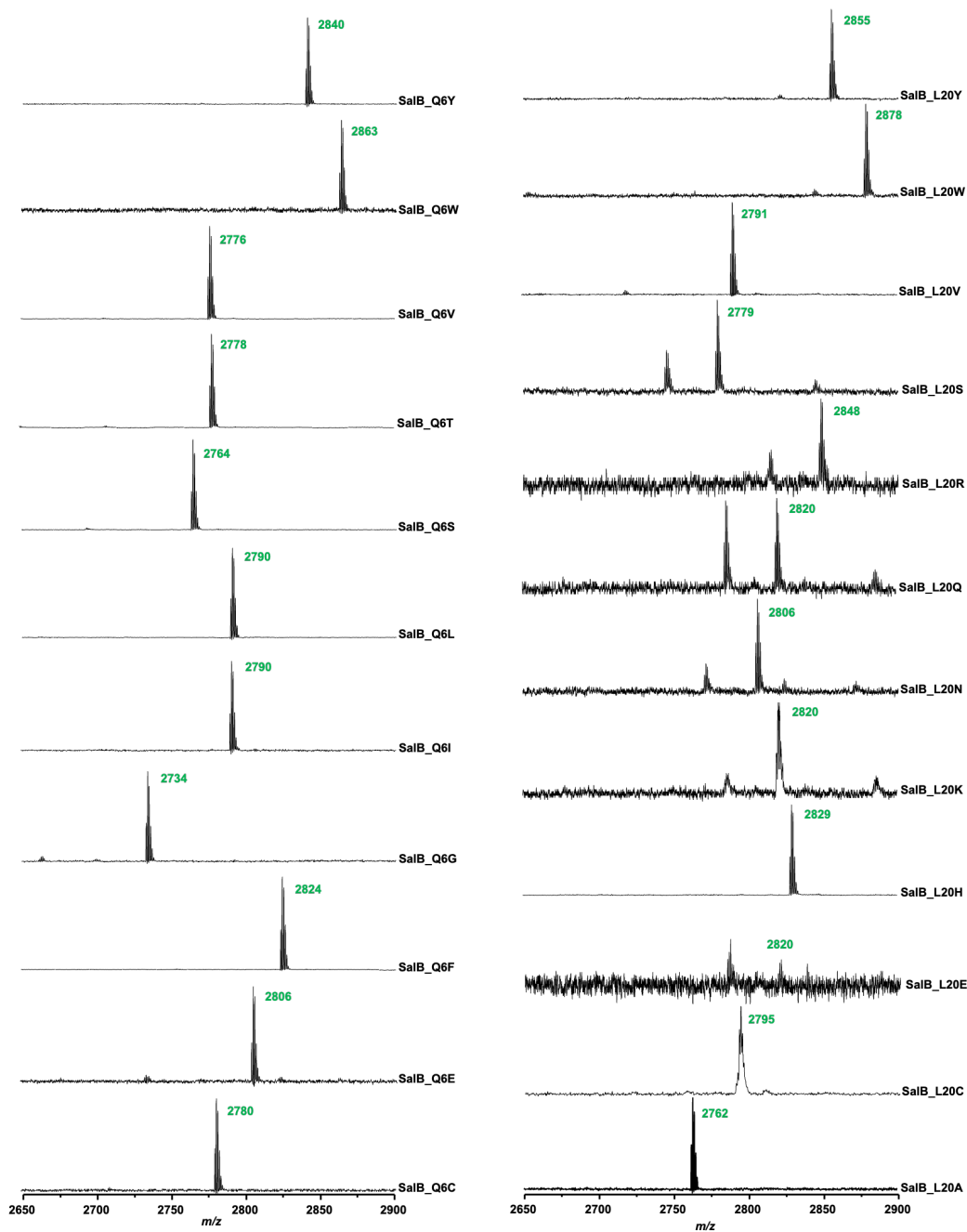
Soluble peptides: SalB_L20C, SalB_L20E, SalB_L20F, SalB_L20H, SalB_L20I, SalB_L20K, SalB_L20M, SalB_L20N, SalB_L20Q, SalB_L20R, SalB_L20S, SalB_L20T, SalB_L20V, SalB_L20W, SalB_L20Y, SalB_Q6A, SalB_Q6C, SalB_Q6E, SalB_Q6F, SalB_Q6G, SalB_Q6H, SalB_Q6L, SalB_Q6M, SalB_Q6N, SalB_Q6S, SalB_Q6T, SalB_Q6V, SalB_Q6W, and SalB_Q6Y. Insoluble peptides: SalB_L20A, SalB_L20D, SalB_Q6I, SalB_Q6K, and SalB_Q6P. Non-expressed peptides: SalB_L20G, SalB_L20P, SalB_Q6D, and SalB_Q6R. Peptides are labeled with the anti-His antibody. Results were reproduced three times independently; representative data are shown. Source data are provided as a Source Data file.



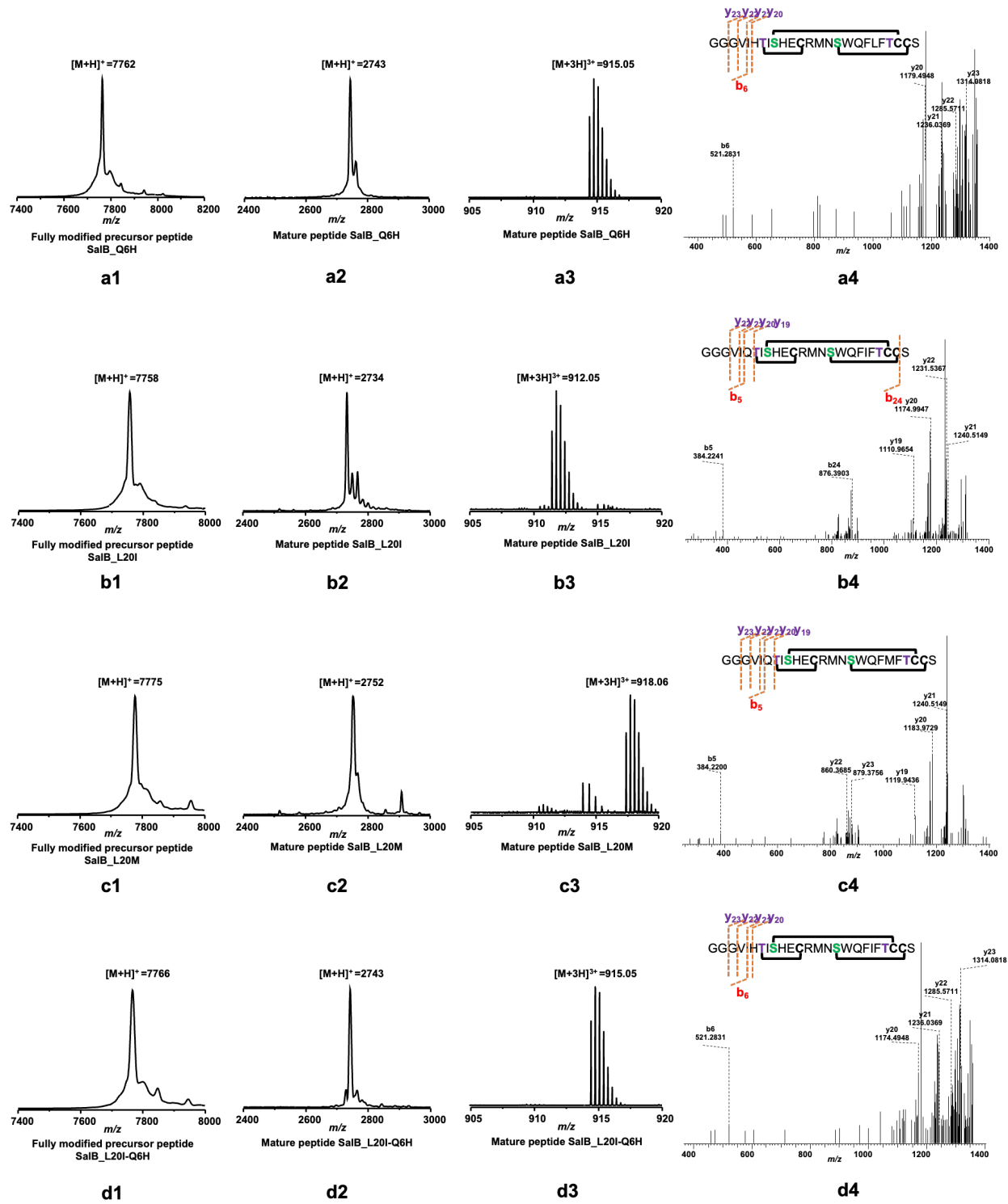
(continued)



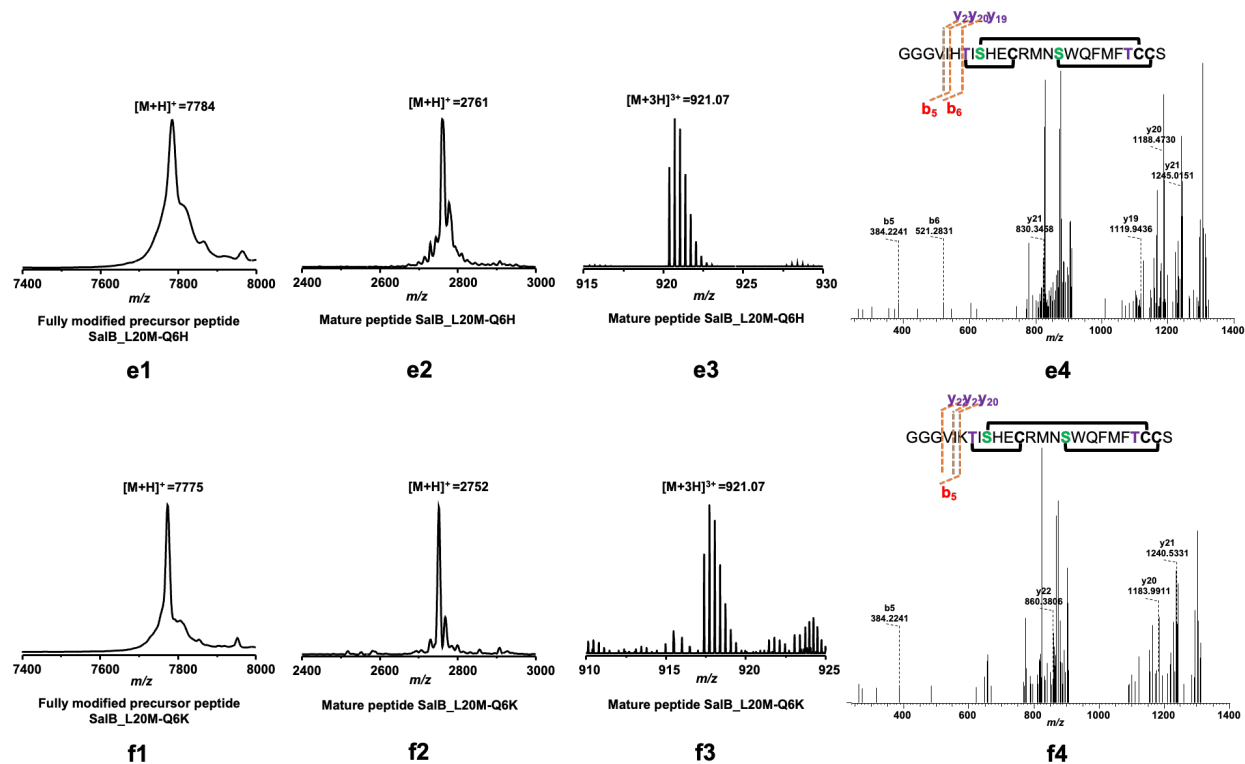
Supplementary Figure 9. MALDI-TOF-MS analysis of cell-free expressed mutated precursor peptides. The products were purified and identified by MALDI-TOF-MS with linear positive mode. The mass labels correspond to the $[M+H]^+$ ion of the synthesized products.



Supplementary Figure 10. MALDI-TOF-MS analysis of salivaricin B mutants (mutated at Q6 or L20). Products were identified by MALDI-TOF-MS with reflectron positive mode. The mass labels correspond to the $[M+H]^+$ ion of the mutant peptides derived from salivaricin B-1 (note that salivaricin B-1 based variants are the most abundant products in all analyses).

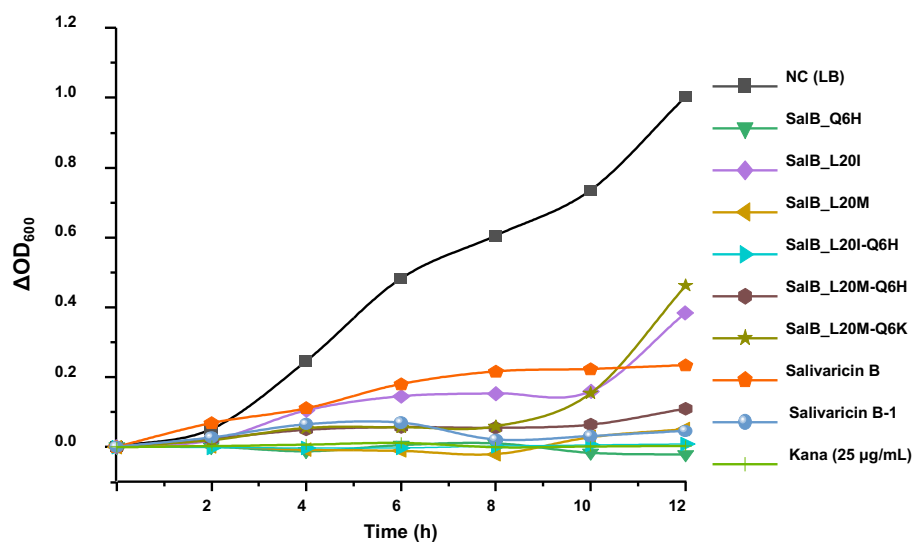


(continued)



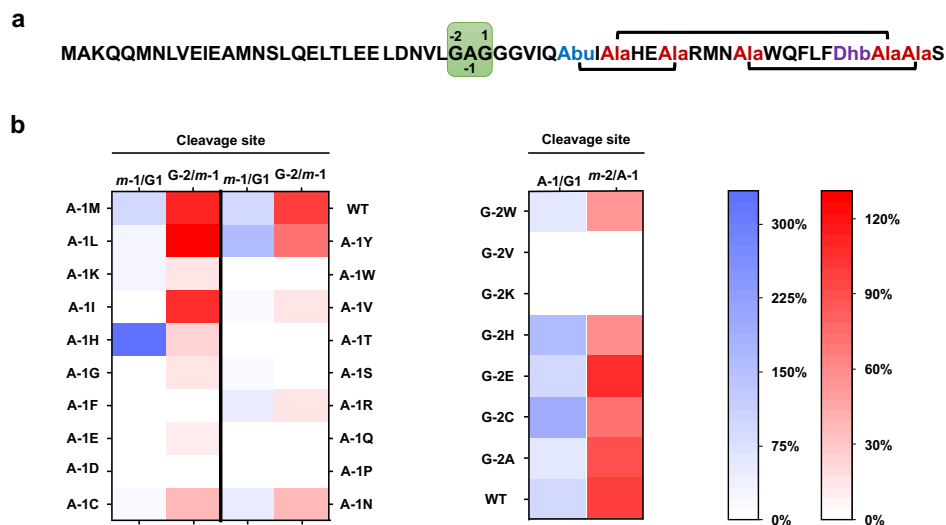
Supplementary Figure 11. Identification of *in vivo* synthesized salivaricin B variants.

(a1-f1) Fully modified precursor peptides identified by MALDI-TOF-MS with linear positive mode. (a2-f2) Mature peptides identified by MALDI-TOF-MS with linear positive mode. (a3-f3) Mature peptides identified by LC-MS. (a4-f4) MS/MS analysis of mature peptides.



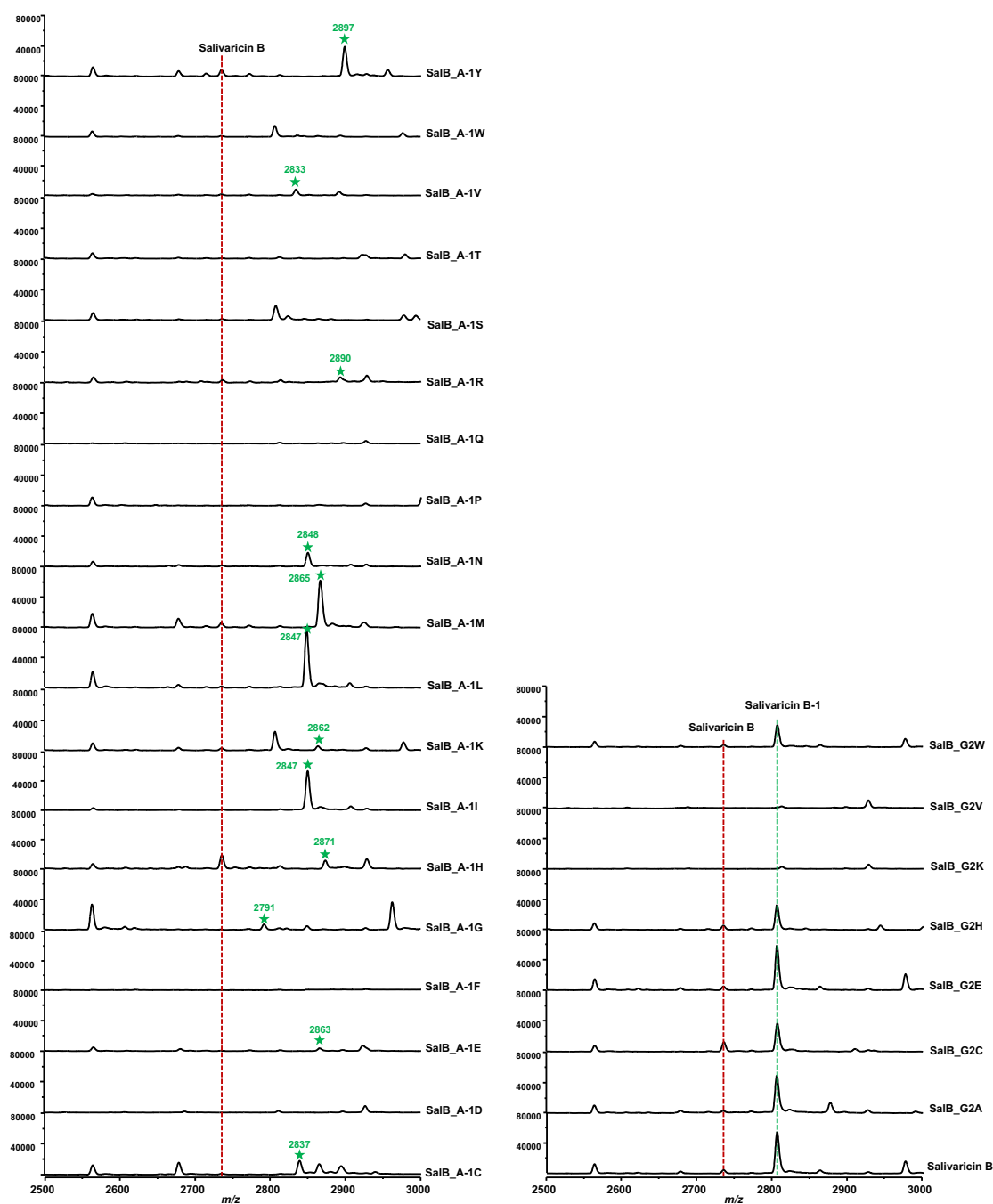
Supplementary Figure 12. Antimicrobial activity assay of selected salivaricin B variants.

Antimicrobial activity assay showing ΔOD_{600} profiles of *S. aureus* RN4220 growth treated with selected single- and two-site mutants. These tested mutants were synthesized and modified *in vivo*, followed by purification and *in vitro* maturation with trypsin digestion. In each activity assay, 50 μM of each pure mutant was added to the cultivation. NC, negative control by adding an equal volume of LB medium to the cultivation. Kanamycin (Kana) is used as a positive control. The data were obtained from one experiment ($n = 1$). Source data are provided as a Source Data file.



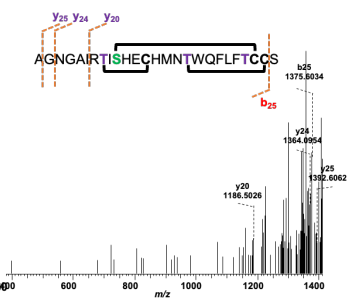
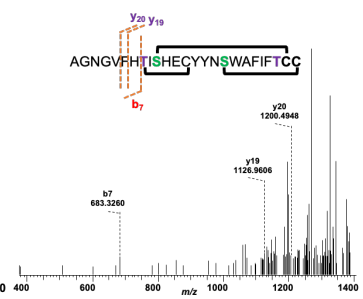
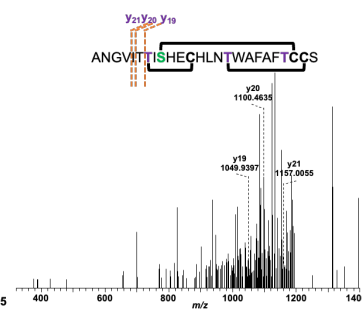
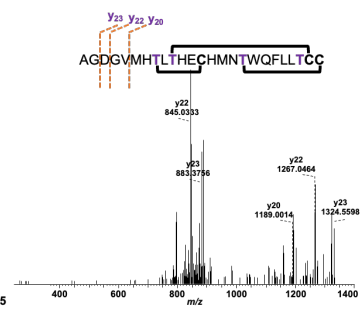
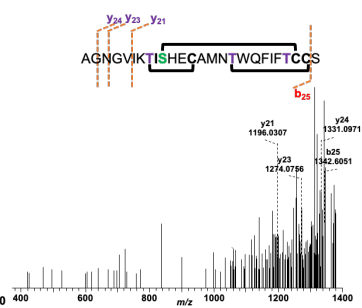
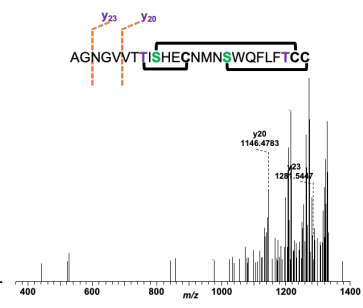
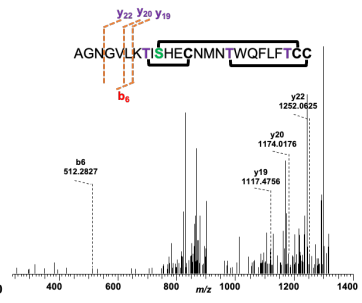
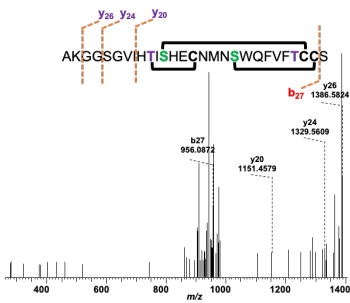
Supplementary Figure 13. Evaluating substrate tolerance of SboT (cleavage site “GAG” is highlighted).

(a) SboM modified precursors (SboA) serving as substrates for SboT. The position of A-1 was selected for saturation mutagenesis. The position of G-2 was mutated to seven other amino acids (i.e., W, V, K, H, E, C, and A). **(b)** Effect of specific-site mutation (A-1 or G-2) on the cleavage site of SboT. Heatmap analysis indicates the relative intensity (%) of mutated products to wild-type peptides (the percent values are calculated according to the mass peaks observed by MALDI-TOF-MS, see **Supplementary Figure 14**). *m*, mutated amino acids at the position of -1 or -2; WT, wild-type SboA without mutation. Source data are provided as a Source Data file.

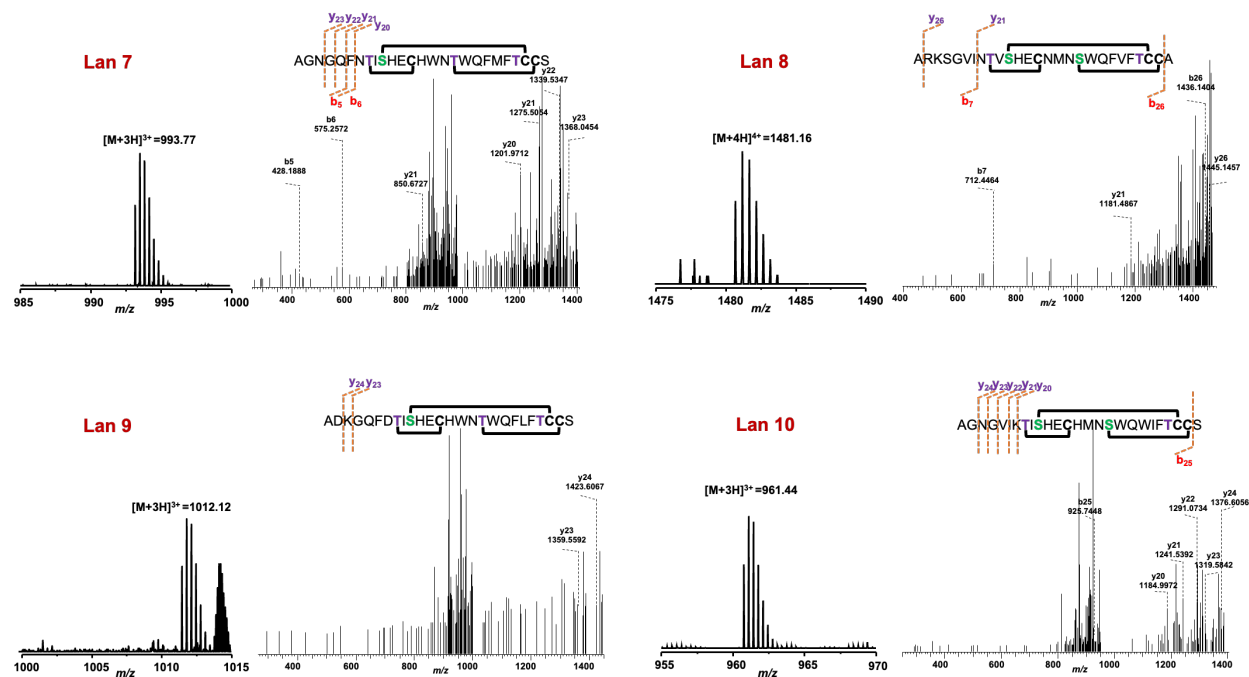


Supplementary Figure 14. MALDI-TOF-MS analysis of peptides matured by SboT.

Products were identified by MALDI-TOF-MS with linear positive mode. Red lines indicate salivaricin B resulted from cleavage site of mutant-1/G and A-1/G. Green stars indicate products resulted from cleavage site of G-2/mutant. Green line indicates salivaricin B-1 resulted from cleavage site of mutant-2/A-1. See **Supplementary Figure 13** for the cleavage site “GAG” of SboT and mutated positions.

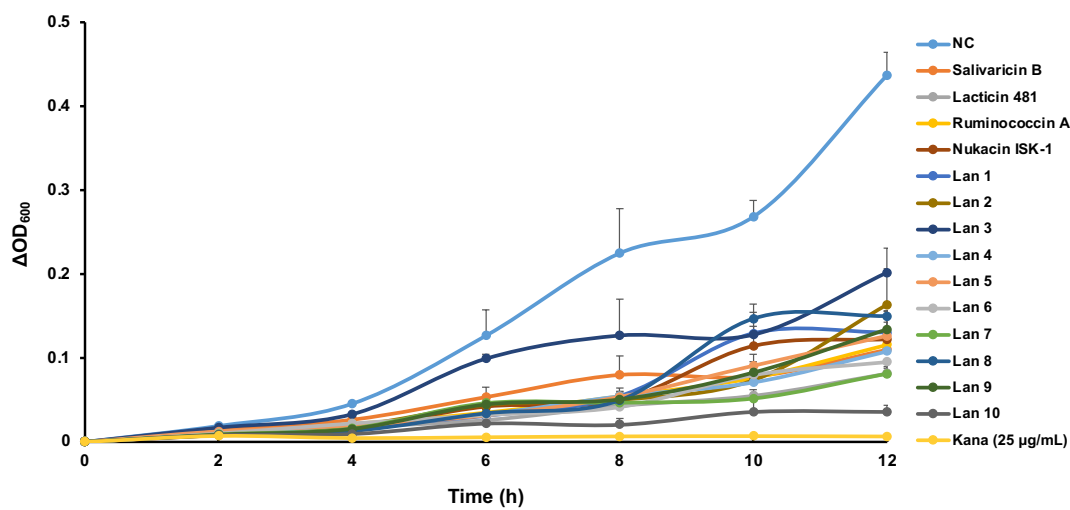


(continued)



Supplementary Figure 15. LC-MS/MS analysis of two known (lacticin 481 and ruminococcin A) and ten uncharacterized lanthipeptides (Lan 1 – Lan 10).

All lanthipeptides were cell-free synthesized in UniBioCat reactions. Note that all sequences are presented with one additional amino acid alanine at the *N*-terminus of the final peptide.



Supplementary Figure 16. Antimicrobial activity assay of ten uncharacterized lanthipeptides (Lan 1 – Lan 10).

Antimicrobial activity assay showing ΔOD_{600} profiles of *S. aureus* RN4220 growth treated with cell-free synthesized novel lanthipeptides. Note that cell-free reaction mixture is used directly for the assay without purification of each lanthipeptide. NC, negative control without gene templates in the cell-free reaction. Kanamycin (Kana) is used as a positive control. Data are presented as mean $\pm$ s.d. of two independent experiments ($n = 2$). Source data are provided as a Source Data file.

Supplementary References

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